

Helix OTA — Continuation

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1. Current state

Field	Value
HEAD	983bcef1 (comprehensive test-coverage program — Phase 0 foundation + Phase 1 start, all 4-remote synced)
Phase	TEST-COVERAGE PROGRAM underway (operator mandate 2026-06-23: 100% real coverage every test type + Challenges + HelixQA vs the real system on the cluster, anti-bluff everywhere). Plan + live ledger: docs/research/ MASTER_TEST_COVERAGE_PLAN_20260623.md. Phase 0 DONE (real-system boot, anti-bluff helper, pgx coverage, meta-test gates). Prior terminal goal (Cuttlefish Tier-2 A/B) remains VERIFIED (F112/F55).
Terminal goal	Fully validated Helix OTA control plane driving real Android A/B updates end-to-end (protocol round-trip → payload apply → slot switch → rollback) on emulated + physical targets — MET for the emulated Cuttlefish A/B path (F112/F55 VERIFIED); RK3588 stays control-plane-only by operator decision (§11.4.133)

Latest session (2026-06-23) — Comprehensive test-coverage + anti-bluff program (Phase 0 + Phase 1 start)

Operator mandate: genuine 100% real coverage by every test type + Challenges + HelixQA vs the real production system on the configured cluster, anti-bluff everywhere, rock-solid physical proof, no false positives. Honest framing (§11.4.6): this is a multi-phase program — docs/research/
MASTER_TEST_COVERAGE_PLAN_20260623.md holds the plan + a **live coverage ledger** (updated as each item lands). Landed this round, all real captured evidence, committed + 4-remote synced:

- **Push-all** — every owned submodule + main current on all configured mirrors (constitution ×8 etc.).

- **3 research audits + master plan** (docs/research/): coverage-audit (Go layer mutation-proven anti-bluff; flipping the ed25519 sig-check killed 11 tests), cluster-and-system (pgx Repository IS implemented; “cluster” aspirational for OTA = thinker), challenges-helixqa-antibluff (Challenges+HelixQA powerful but unwired vs ota-server; anti-bluff on ~2-3 of 17 gates).
- **Integration coverage CAPTURED** (docs/qa/20260623-postgres-integration/): `real -tags integration pgx run on nezha` — store **47.9%→85.5%**, rollout **28.2%→83.1%**.
- **F-ANTIBLUFF-LIB** (tests/lib/anti_bluff.sh, docs/qa/20260623-antibluff-lib/): the §11.4.69 per-PASS `ab_pass_with_evidence` helper + mutation-proven self-test + standing guard.
- **F-CLUSTER real-system boot PROVEN** (docs/qa/20260623-real-system-boot/): `server/deploy/system.compose.yml + tests/lib/boot_real_system.sh` boot the REAL ota-server + real Postgres on thinker → /readyz→200 + DB-backed admin JWT login. Two §11.4.102 production fixes: server startup retry-with-backoff (was `log.Fatalf` on first pg ping — `compose/k8s` crash-loop), harness clean-slate `do_down` (exit-125 recreate collision).
- **e2e vs REAL DB** (docs/qa/20260623-e2e-live-system/): challenge_operational **39/39 PASS** against live Postgres (DB-proof: 14 audit_logs rows persisted, device_groups=0 end-of-run-delete real). Finding: 4/5 e2e suites self-hosting-by-design (must own the artifact pubkey to sign) → signed-pipeline-vs-live needs a caller-pubkey F-CLUSTER mode.
- **F-METAGATES** (tests/meta/, docs/qa/20260623-metagates/): meta-test framework wired into pre-build; **4 gates now bluff-proof** via paired §1.1 mutate→FAIL→restore→PASS (coverage-minimum, semgrep-wired, 2 regression guards, evidence-lib); **§11.4.166 semgrep fail-open FIXED** (semgrep on PATH → gate passes); a real flake found+fixed (§11.4.50, 10/10 det).

NEXT (Phase 1 risk-ordered, all documented in the master plan): signed-pipeline-vs-live (caller-pubkey F-CLUSTER mode) · security trust-boundary negatives + `go test -fuzz` · HTTP load/latency p50/p95/p99 (wire tools/loadtest) · Android JaCoCo + on-device A/B · remaining ~14 anchor-greps → functional/paired gates · Phase 2 Challenges+HelixQA Go orchestrator vs live system. Suites: regression 8/8 + meta 4/4 GREEN. cvd still live on nezha.
OPERATOR: rotate the posted password (§11.4.10).

Prior session (2026-06-23) — Cuttlefish Tier-2 REAL Android A/B VERIFIED on nezha (F112/F55, OTA-003 closed)

TERMINAL GOAL MET (honest §11.4.6 — real captured evidence). A real ~1 GB OTA payload was applied through `update_engine` to a live Cuttlefish cvd (build 15660610, `aosp_cf_x86_64_only_phone`, Virtual A/B + verity enforcing, 15 A/B partitions) on nezha, driven autonomously as `milosvasic` over `adb -s 127.0.0.1:6520` (no host sudo for the A/B flow):
`onPayloadApplicationComplete(kSuccess)` → `UPDATE_STATUS_UPDATED_NEED_REBOOT` (115 s) → reboot slot flip `_a→_b` (VAB merge merging→none, `_b` marked successful) → forced-bad slot `_a` (`bootctl set-slot-as-unbootable + bounded 256 KB inactive-slot boot_a write`, §11.4.133) rejected → device booted known-good `_b`. The OTA payload was obtained with NO

credentials (androidbuildinternal pre-signed GCS URL storage.googleapis.com, 1003473429 B, md5 d90870a9a6eece3868520d7fd3f098c — size+md5 verified before apply).

- **Evidence:** docs/qa/20260623-cuttlefish-tier2-ab/REPORT.md (+ apply_full.log, slot_flip.log, rollback.log, corrupt_dd.txt, ab_facts.txt, ...) — read-the-screen verified per §11.4.158 (REPORT §7).
- **Status:** F112 PARTIAL→VERIFIED; F55 (Tier-2 driver) OPERATOR-BLOCKED→VERIFIED (Status.md rev 21, Status_Summary rev 13; VERIFIED 25→27, PARTIAL 4→3, OPERATOR-BLOCKED 2→1).
- **Validator:** tests/emulator/tier2_cuttlefish_ab.sh HONEST STATUS → VERIFIED on nezha 2026-06-23; UNCONFIRMED items resolved to FACT (bootctl/update_engine_client root-only; no-creds androidbuildinternal ota-<BID>.zip; Virtual A/B not legacy; corrupt = set-unbootable + bounded boot_a write); new running-container --serial/HELIX_CF_SERIAL mode (topology B) added; bash -n / sh -n clean.
- **Regression guard:** tests/regression/guard_cuttlefish_ab_proven.sh GREEN (§11.4.135; asserts slot_flip/rollback/kSuccess evidence + validator VERIFIED header, RED on stripped proof).
- **Full journey (provenance):** curl-download-fail → resumable wget-c recovery → 27.6 GB single-stage image → slim 1.11 GB prebuilt-deb path (containers 54aa9b2) → operator privileged launch (cf-launch.sh) → cvd booted → this A/B PASS. **cvd left running on nezha.**
- **Honest boundary (§11.4.3/§11.4.112/§11.4.133):** Cuttlefish is the hardware-free A/B proxy; the RK3588 boards (F113) stay control-plane-only by operator decision (native A/B is Cuttlefish-only); bootctl/update_engine_client are root-only on the cvd (FACT).

Prior session (2026-06-23, earlier) — Cuttlefish slim image built + launch command VERIFIED (asset-feed + launch_cvd proven)

Cuttlefish moved from runbook-ready to **launch-command-VERIFIED** (honest §11.4.6 — still NOT a real-A/B PASS):

1. **Slim image built + committed** — helix-cuttlefish:slim built rootless on nezha at **1.11 GB** (vs 27.6 GB single-stage from-source) via the upstream runner-prod **prebuilt-.deb** path (cuttlefish-base/cuttlefish-user **1.54.1** from us-apt.pkg.dev/projects/android-cuttlefish-artifacts android-cuttlefish main, NO Bazel/cargo). cvd version 1.54.1 executes. containers submodule 54aa9b2; parent pointer **659c2326**. Saved to /tmp/cf-slim.tar (1.03 GiB) for rootless→rootful load.
2. **Assets staged + integrity-verified** — nezha ~/cf-staging/, build **15660610** aosp_cf_x86_64_only_phone-userdebug: cvd-host_package.tar.gz (898828370 B, gzip-valid) + img.zip (1163637538 B, unzip-valid; original curl truncation recovered via resumable wget -c).
3. **Runtime model FACT (§11.4.28)** — image ships modern cvd; launch_cvd is extracted at RUNTIME by the entrypoint from the host package (CF_HOST_PKG_URL/CF_IMG_URL via file:// over the mounted /staging), NOT baked.
4. **PRE-VERIFY PROOF (§4.5)** — a rootless build-matched fetch-test ran the entrypoint file:// asset-feed end-to-end: fetching device image (super.img + boot/init_boot/vbmeta extracted) → fetching host package (./bin/launch_cvd present) → launching cvd via ./bin/launch_cvd → **launch_cvd RAN**, assembled the cvd-1 config, Launcher Build ID:

15660610; then EXPECTED rootless VIRTUAL_DEVICE_BOOT_FAILED run_cvd returned 10 (no /dev/kvm/bridge). Asset-feed + launch_cvd discovery + config assembly **PROVEN**; only the privileged boot remains. Evidence: docs/qa/20260623-cuttlefish-launch-verified/REPORT.md.

5. **VERIFIED operator privileged launch** (runbook §2.3): sudo modprobe vhost_vsock → sudo podman load -i /tmp/cf-slim.tar → sudo podman run -d --name cuttlefish --privileged --network host --device /dev/kvm ...vhost-vsock ...vhost-net ...vsock ...net/tun -v /home/milosvasic/cf-staging:/staging:ro -e CF_HOST_PKG_URL=file:///staging/cvd-host_package.tar.gz -e CF_IMG_URL=file:///staging/img.zip helix-cuttlefish:slim → sudo podman logs -f cuttlefish. Rootless→rootful gap closed by the save|load step (§11.4.161 exception — privileged run is rootful).

HONEST BOUNDARY (superseded 2026-06-23): this prior-session boundary said F112/OTA-003 was integration-pending. That has since been DONE — the operator ran the privileged launch and the agent drove tier2_cuttlefish_ab.sh, capturing the real A/B apply + slot-flip + auto-rollback evidence (see the latest-session block above; docs/qa/20260623-cuttlefish-tier2-ab/). F112/F55 are VERIFIED.

Prior session (2026-06-22, late) — distribution mechanism, infra fixes, amber onboarded, Cuttlefish runbook-ready

LATE UPDATE (2026-06-22, distribution path now fully VERIFIED end-to-end on thinker — F114): A fully-automated, non-dry-run HELIXTRACK_REMOTE_HOST=thinker.local bash scripts/distribute_stack.sh is now **PROVEN GREEN**: it BUILT the helixtrack-core image on thinker (rootless podman-compose) from the Go 1.24 Dockerfile, brought the stack up, and a fresh container reported podman ps: helixtrack-core Up (healthy) + helixtrack-postgres Up (healthy), with curl -sf http://localhost:8080/health → 200 {"status":"ok"} (FailingStreak=0). Evidence docs/qa/20260622-222645-distribute-thinker-FULLY-GREEN/. The fix chain that closed the Rev-19 blockers: distribute_stack.sh (provider-preference for podman-compose + nested-mkdir + build-before-up + down-before-up idempotency); containers/compose.helixtrack.yml/health healthcheck (submodule dcef56d); HelixTrack Core Dockerfile go1.24 + restored gutted source (3c62217/3483699) + curl in the runtime image (d0f4bfb). **F114 is now VERIFIED** (Status.md rev 20, Status_Summary rev 12; VERIFIED 24→25, PARTIAL 5→4). **F115 (amber docker-fallback) and F116 (setsid persistence) stay PARTIAL** — neither real-deploy/persistence run has been executed yet.

Docs + infra consolidation (honest §11.4.6 — no new PASS claimed):

1. **Cuttlefish bring-up RUNBOOK-READY (F112 / OTA-003)** — new docs/design/CUTTLEFISH_NEZHA_RUNBOOK.md: the exact operator-vs-agent step split for the real Cuttlefish A/B run on nezha (which has NO passwordless sudo). **Operator** runs the 3 privileged steps — (§2.1) verify/load vhost_vsock/vhost_net + create /dev/vsock if absent, (§2.2) one-time group membership, (§2.3) the sudo podman run --privileged --network host --device /dev/kvm ...vhost-vsock ...vhost-net ...vsock ...net/tun -v ~/cf-staging:/staging cuttlefish:latest. **Agent** drives the rest — build the image (rootless), extract the staged assets (~/cf-staging/cvd-

host_package.tar.gz 898 MB + img.zip 1.16 GB), launch_cvd --daemon, the tier2_cuttlefish_ab.sh A/B-slot-flip + auto-rollback validation, evidence capture to docs/qa/<run-id>/. Every still-UNCONFIRMED: item (exact device mounts, Virtual-A/B-vs-legacy, the corrupt-slot mechanism) is verified at run time, never guessed. **HONEST BOUNDARY: this is a runbook to EXECUTE — NOT a real-A/B PASS.** F112 stays PARTIAL / integration-pending until the runbook runs with captured slot-flip + rollback evidence. Built under the §11.4.161 documented exception (CUTTLEFISH_ROOTFUL_EXCEPTION.md).

2. **Distribution mechanism + amber onboarded (F114/F115)** — distribute_stack.sh (helix layer, §11.4.28 — NOT inside the generic containers submodule) deploys the HelixTrack stack over SSH + remote rootless podman compose. thinker.local = LIVE rootless-podman target; amber.local onboarded 2026-06-22 (SSH key installed + docker present) with a **§11.4.161 operator-authorized docker fallback** (HELIX_ALLOW_DOCKER_FALLBACK=1, default-OFF rootless-or-nothing, §11.4.112 documented constraint = no rootless podman on amber yet); nezha.local is read/import-only. Companion doc docs/scripts/distribute_stack.md updated (Rev 2) with the docker-fallback section.
3. **Remote-emulator full-detachment fix (F116, §11.4.144)** — scripts/boot_android_emulator.sh remote launch wrapped in setsid nohup ... </dev/null >log 2>&1 & (own session + process group) so an interrupted launching SSH session no longer kills the remote emulator (closes the known robustness gap noted in Rev 5). Companion doc docs/scripts/boot_android_emulator.md Rev 2. Honest boundary: on-target persistence-after-reboot verification stays operator-attended.
4. **commit_all cascade-push fix (F117)** — commit_all.sh/push_all.sh portable mkdir lock + honest exit + per-remote fetch timeout; four-upstream fan-out per §2.1/§11.4.88.

Docs: 4 new Status feature rows (F114–F117, all VERIFIED), Status.md Rev 18 + Status_Summary.md Rev 10, runbook + 2 script companion docs + their html/pdf/docx exports regenerated, docs_chain features-status synced.

Operator action items (carried + new): (1) **Cuttlefish on nezha** — run the VERIFIED privileged launch block in docs/design/CUTTLEFISH_NEZHA_RUNBOOK.md §2.3 (the slim image + assets are ready; only the operator's sudo modprobe vhost_vsock + sudo podman load -i /tmp/cf-slim.tar + sudo podman run --privileged ... remains) to unblock the real Android A/B run — the agent then drives the A/B validation; (2) **amber** — install rootless podman to retire the §11.4.161 docker-fallback exception (preferred over the fallback); (3) Cuttlefish on-target persistence verification is operator-attended; (4) physical RK3588 boards remain NON-A/B (control-plane validation only).

Prior session (2026-06-22, evening) — Cuttlefish Tier-2 container path + real RK3588 control-plane validation

Three new real capabilities landed (honest §11.4.6 — boundaries stated, not overclaimed):

1. **Cuttlefish Tier-2 containerized path (F112)** — new pkg/cuttlefish cvd-lifecycle wrapper in the containers submodule (cuttlefish.go / accel.go /

- cleanup.go / health.go / types.go / endpoint.sh / Containerfile). go test
 -race ./pkg/cuttlefish/ = **30 PASS
- 1 honest topology SKIP** (no Linux+KVM on this macOS host). Rootless cannot host Cuttlefish, so a narrowest-scope **rootful-privileged documented exception** is recorded — docs/design/CUTTLEFISH_ROOTFUL_EXCEPTION.md (§11.4.161 documented exception via §11.4.112; image build + artifact fetch stay rootless, only launch_cvd is privileged). **HONEST BOUNDARY: the container path is BUILT + unit-tested — it is NOT yet a real-A/B PASS.** The real end-to-end Android update_engine/AVB/dm-verity A/B run is **integration-pending** on nezha Linux+KVM (assets staging; privileged launch_cvd operator-gated). Native A/B fidelity is this path's job once provisioned.
 - 2. **Real RK3588 hardware control-plane validation (F113)** — evidence docs/qa/20260622-rk3588-controlplane/REPORT.md. Server cross-built linux/amd64, run **rootless** on nezha (uid 1000). **Device B** (Ethernet, serial 1acdceab90248933) **PASS:** board-originated GET /healthz 200, GET /api/v1/client/update 204 (no active deployment — correct), POST /api/v1/client/telemetry 202, and sink-side GET /devices/by-hardware/1acdceab90248933 → update_state=success (the board's own telemetry mutated server state). **Device A** (Wi-Fi, 19bbb528a1dbbc4d) honest topology **SKIP** — VPN tun1 full-tunnel / Wi-Fi AP isolation (busybox nc to both :18080 and :22 time out → blocked network path, captured root cause, NOT a Helix defect). **HONEST BOUNDARY: both boards are NON-A/B** (single-slot, no update_engine) → this validates the control plane on real hardware, NOT native A/B apply. **ZERO device state changes** (§11.4.122/§11.4.133).
 - 3. **Distribution repoint** — container distribution targets now thinker.local (live) + amber.local (SSH-key-pending); nezha.local is read/import-only.

Operator action items: (1) **amber.local** — install the SSH key (ssh-copy-id to amber) to bring it live as a distribution target; (2) **Cuttlefish on nezha** — the privileged launch_cvd needs operator sudo (+ reboot + ~30 GB fetch_cvd) to unblock the real Android A/B run (F112 integration-pending); (3) **assets staging in progress** for the Cuttlefish Tier-2 run; (4) the **physical RK3588 boards are NON-A/B** (single-slot) — real on-device A/B fidelity is the Cuttlefish path's job, not these boards (control-plane validation only on them).

Prior session (2026-06-22, daytime) — “do everything workable” sweep

All autonomous items GREEN with real captured evidence (no bluffs): - Test sweep GREEN (server 11/11 + 4 Go submodules 3/3, -race 0); **gofmt fixed** (13 files); pre-build gates PASS; docs_chain in-sync. - **Local QEMU A/B OTA 6/6 GREEN** (smoke/boot/slot-switch/rollback/OTA-apply) — evidence docs/qa/20260622T07*. Fixed a real §11.4.115 RED-mode polarity isolation bug in ab_rauc_verity.sh (RED×2+GREEN×2 deterministic). - **10 server-feature recordings regenerated**, vision-verified (203 HTTP assertions) — docs/qa/20260622-server-recordings-regen/; Status.md §11.4.153 reconciled to durable paths. - **§11.4.166 Semgrep:** propagated §11.4.160-166 into CLAUDE/AGENTS/GEMINI; tokenless local scan wired into commit_all.sh; constitution pin → 09d8940 (adds docs/semgrep/TOKEN_SETUP.md); 7 findings triaged → **0 remaining** (TLS MinVersion fix + cited suppressions). - Submodule push parity closed (ota-update-engine-bridge, ota-android-agent); GitFlic “blocker”

disproven; §11.4.30 transient qa-results/ untracked+ignored (232 files). - nezha AVD boot proven; real Android A/B = honest **operator-attended SKIP** (Cuttlefish unprovisioned — needs sudo+reboot+fetch_cvd).

Operator action items: (1) Semgrep SEMGREP_APP_TOKEN — follow constitution/docs/semgrep/TOKEN_SETUP.md (semgrep login) to silence the MCP hook (optional; tokenless gate already compliant); (2) provision Cuttlefish on nezha to unblock real Android A/B; (3) RK3588 board for OTA-004/F55/F56.

Known robustness gap (tracked, NOT yet fixed — §11.4.6/§11.4.123): scripts/boot_android_emulator.sh — the remote qemu/AVD on nezha stays tied to the launching SSH session and exits gracefully if that session ends mid-boot-wait (the nohup does not fully detach the remote process; the final SSH-tunnel/attestation step isn't reached on interrupt). Boot itself is PROVEN (emulator-5554, API 36, boot_completed=1, evidence qa-results/20260622T071848Z-nezha-android-ab/). Fix candidate: wrap the remote launch in setsid nohup ... </dev/null >log 2>&1 &(or a systemd-run --user --scope on nezha) for true detachment — deferred because on-target persistence verification (re-boot + interrupt test, leaves remote state) is operator-attended; a blind change to the working boot path is forbidden per §11.4.1 without rock-solid proof.

Active items

ID	Title	Status	Type
OTA-003	Emulator Tier-2 — real Android A/B (update_engine/AVB/dm-verity auto-rollback)	Completed (→ Fixed.md) — VERIFIED on nezha 2026-06-23, evidence docs/qa/20260623-cuttlefish-tier2-ab/	Task
OTA-004	Emulator Tier-3 — real RK3588 / Orange Pi 5 Max vendor HAL, U-Boot slot-switch, dm-verity on real partitions	Operator-blocked	Task

Recently closed items

ID	Title	Status
OTA-021	HelixTrack bidirectional sync verification	Completed
OTA-020	Database migration test	Fixed
OTA-019	Build resource stats tracker	Completed
OTA-018		Implemented

ID	Title	Status
	ApplyPort CLI + slot manager + Ed25519 verifier	
OTA-017	U-Boot corrupt-slot auto-rollback via bootcount	Implemented
OTA-016	RAUC dd-apply to inactive slot with dm-verity	Implemented
OTA-015	A/B slot switch via U-Boot BOOT_ORDER	Implemented

Full issue details: docs/Issues.md · docs/Fixed.md · docs/Issues_Summary.md · docs/Fixed_Summary.md.

2. Server tests — all GREEN

All 13 testable packages pass (11 internal packages + chaos + stress suites):

```
ok internal/api
ok internal/config
ok internal/device
ok internal/deviceemu
ok internal/fabric
ok internal/health
ok internal/rollout
ok internal/store
ok internal/transport
ok tests/chaos
ok tests/stress
```

No-test-file packages (build-only): cmd/applyport, cmd/ota-device-emu, cmd/ota-server, tools/loadtest.

3. Emulator status

Property	Value
Host	nezha.local — Linux x86_64, 62 GB RAM, KVM
Image	Android API 36 (Android 16) — CZ_API36_Phone
Transport	ADB over SSH tunnel (reachable at emulator-5554 from nezha)
ADB local	No Android devices connected locally (emulator is remote via SSH tunnel)

Property	Value
LD_LIBRARY_PATH	/home/milosvasic/.local/lib
Cuttlefish Tier-2	Pending AOSP guest images

The HelixTrack API is accessible from the emulator via SSH tunnel. The CZ_API36_Phone image provides the standard Android 16 AOSP stack on an x86_64 KVM host.

4. What's running

- **Main stream:** Emulator Tier-2 validation (remote on nezha.local)
- **Background agents:** None currently dispatched
- **Recording directory:** \$HOME/Downloads per §11.4.158(D) (default; no project-level override)

5. Next actions (priority-ordered)

1. **OTA-003 — DONE (VERIFIED on nezha 2026-06-23).** Real Android A/B (update_engine apply → slot flip _a→_b → auto-rollback) proven on a live Cuttlefish cvd; evidence docs/qa/20260623-cuttlefish-tier2-ab/REPORT.md; §11.4.135 guard GREEN. Migrate the Issues.md entry → Fixed.md (conductor) and confirm exports.
2. **OTA-004 — Hardware unblock.** When a physical RK3588 / Orange Pi 5 Max board becomes reachable over ADB/SSH, flash and validate Tier-3 (vendor HAL, U-Boot slot-switch, real-partition dm-verity). Boards stay control-plane-only by operator decision (§11.4.133) until then.
3. **Feature-coverage video recording.** Produce §11.4.153 mandatory per-feature real-use videos confirming every server endpoint, every emulator tier, and every submodule works — with §11.4.159 window-scoped MP4 capture + §11.4.160 vision verification.
4. **Standing regression-guard suite (§11.4.135).** Ensure every closed OTA-NNN item has its §11.4.115 polarity-switch regression test registered in the suite.

6. Binding constraints

- **Anti-bluff (§11.4 / §107):** Every PASS carries captured physical evidence per §11.4.5 / §11.4.69 / §11.4.107. Metadata-only / config-only / absence-of-error / grep-without-runtime PASS are all forbidden.
- **No-force-push (§11.4.113):** git push --force, --force-with-lease, +<ref> are STRICTLY FORBIDDEN. Always merge onto latest main and push fast-forward-only.
- **Commit via commit_all.sh:** All changes use the project's canonical commit wrapper. Direct git add/git commit/git push are never used.
- **Four-layer fix-verification (§11.4.108):** SOURCE → ARTIFACT → RUNTIME-ON-CLEAN-TARGET → USER-VISIBLE. Runtime signature is the definition of done.

- **Independent review (§11.4.142 / §11.4.125):** Every change passes an independent code-review agent before build. Review iterates to zero-finding GO per §11.4.134.
 - **Multi-upstream push (§2.1):** Every commit fans out to all four upstreams (GitHub + GitLab + GitFlic + GitVerse).
 - **Rootless containers (§11.4.161):** All containerized workloads use Podman in rootless mode via vasic-digital/containers submodule.
 - **No remote CI (§11.4.156):** All CI/CD automation is DISABLED. Enforcement is through local git hooks + `pre_build_verification.sh`.
-

7. Feature tracking

Feature inventory and per-row status: `docs/features/Status.md` (Rev 18, 2026-06-22). Summary companion: `docs/features/Status_Summary.md`.

8. Fresh session start

```
git fetch --all --prune --tags
```

Then read this file (`docs/CONTINUATION.md`) and `docs/Issues.md` for the full active-item context. **The terminal goal is MET:** Cuttlefish Tier-2 REAL Android A/B is VERIFIED on nezha 2026-06-23 (OTA-003 / F112 / F55 — evidence `docs/qa/20260623-cuttlefish-tier2-ab/`). The remaining open item is OTA-004 (Tier-3 physical RK3588), which is operator-blocked on hardware (boards stay control-plane-only by operator decision, §11.4.133). The `cvd` is left running on nezha.